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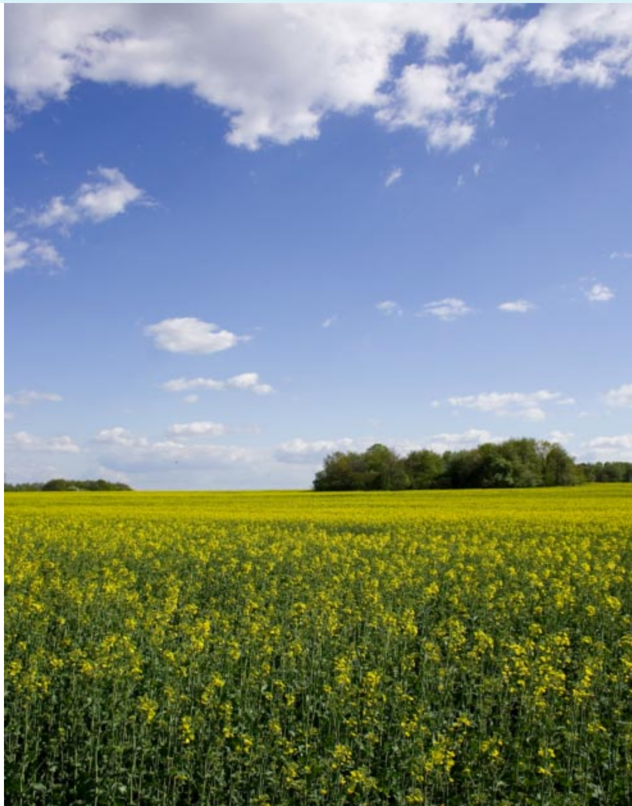
Using OPeNDAP-enabled Applications to Access Australian Data Services and Repositories

eResearch Australasia 2011, ½ Day Morning Workshop, Thursday 10th November 2011





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Thank you

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GENERAL INFORMATION

- A hands-on component is provided, however attendees are not required to participate. Material and Data Sources provided.

- Content found at
  - http://godae.bom.gov.au/eresearch_2011/

- OPeNDAP servers located at
  - Workshop (http://192.168.1.1:8090/opendap)
  - Bureau of Meteorology (http://opendap.bom.gov.au:8080/thredds)
  - CSIRO (http://opendap.csiro.au/thredds)
  - ANU/NCI (http://opendap.nci.org/thredds)
  - OPeNDAP, inc (http://test.opendap.org/opendap)

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WORKSHOP DESCRIPTION

- This half-day workshop will afford attendees an opportunity to discover and access the latest OPeNDAP server systems known as Hyrax and THREDDS in Australian research sites.
- Workshop segments are organized to encourage participants to interact with these data services, so to understand how the services features are utilized by service providers and users with OPeNDAP-enabled applications and tools.
- The workshop will cover four areas, one to discover data services and data products, another to show applicable uses of data services and client applications, another to focus on meta-data discovery and data access/extraction, and a final segment on accessing geospatial and aggregation data services.
- The presenters seek attendee feedback on the utility of the services and client-side tools, and discussions on future tools and applications.

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GENERAL INFORMATION

- This is a half-day workshop (9am to 12:30pm)
  - 9:00am Introductions and Participants Goals
  - 9:15am Session 1: Discovering OPeNDAP data access services
  - 10:00am Session 2: Applicable use cases of OPeNDAP data services
    - 10:30am Tea Break for 15 minutes
  - 11:00am Session 3: OPeNDAP service protocols and features
  - 11:45am Session 4: Accessing complementary features and services
  - 12:30pm End of Workshop

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Workshop Session Outline

First topic (45 minutes)

Discovering OPeNDAP data access services and the features available to client applications.

A short tutorial on exploring and accessing data on a Australian OPeNDAP service with a web browser.

Second topic (45 minutes)

Applicable use cases of OPeNDAP data services for data access using a variety of applications and tools.

A short tutorial exploring data access using an OPeNDAP-enabled tool within a scripting language such as python.

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Workshop Session Outline

Third topic (45 minutes)

OPeNDAP service protocols and features to support meta-data discovery and efficient data access and extraction.

A short tutorial on meta-data and data access and extraction using an OPeNDAP-enable client application.

Fourth topic (45 minutes)

Accessing OPeNDAP servers with geospatial services, aggregation services and virtual datasets.

A short tutorial on data access to these services using an OPeNDAP- and OGC-enabled client application or web servlet.

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Session 1

- Discovering OPeNDAP data access services and the features available to client applications.
- A short tutorial on exploring and accessing data on a Australian OPeNDAP service with a web browser.
- *45 minutes* in length

BROAD VISION

1. A world in which a single ***data access protocol*** is used for the exchange of data between network-based applications regardless of discipline.

2. A layer above TCP/IP providing for syntactic and semantic consistency not available in existing protocols such as FTP.

Fundamental Objective of OPENDAP

The fundamental objective of OPeNDAP and OPeNDAP Inc. is to facilitate internet access to scientific data

This is done by:

- Providing a protocol (DAP) to access data over the internet,
- Hiding the format (and organization) in which the data are stored from the user, and
- Providing subsetting (and other) capabilities for the data at the server

OPeNDAP is based on a multi-tier architecture

OPeNDAP software is open source

Some Definitions

DAP = Data Access Protocol

- ➢ Model used to describe the data;
- ➢ Request syntax and semantics; and
- ➢ Response syntax and semantics.
  - ➢ The data structure returned to the user

OPeNDAP

- ➢ The software that forms the service;
- ➢ Numerous implementations (Hyrax (reference), THREDDS,…);
- ➢ Core/libraries for client applications and services.

THREDDS/Hyrax

- ➢ A service framework (portal) that contains the OPeNDAP service;

Some Definitions

Syntax

- The computer representation of a data object - the data types and structures at the computer level; e.g.,
- T is a floating point array of 20 by 40 elements.

Semantics

- The information about the contents of an object; e.g.,
- T is sea surface temperature in degrees Celsius for a certain region of the Earth.

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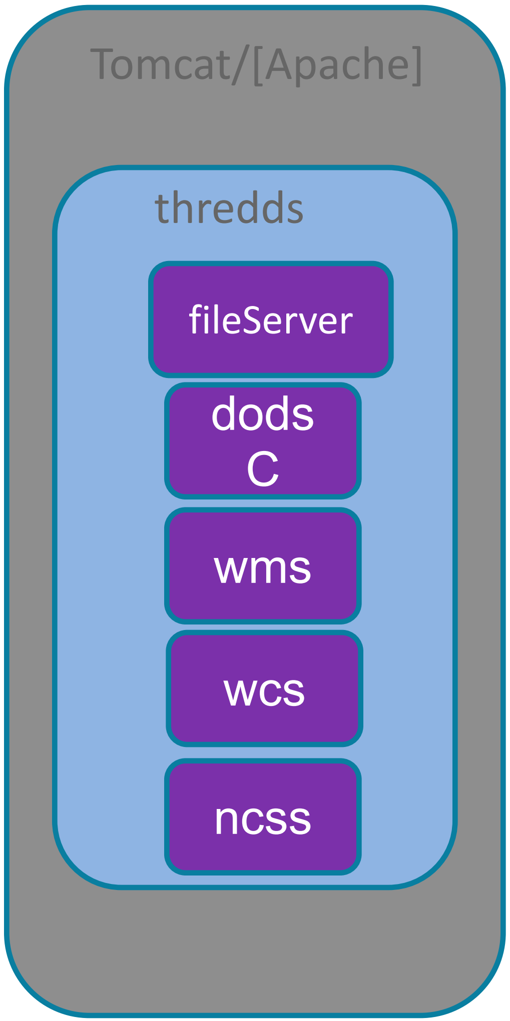
THREDDS services syntax



http://{server:port}/{contextPath}/{service}/...

{contextPath} = "thredds" (servlet default name)
{service} = "fileServer" | "dodsC" | "wms" | "wcs"

- Bulk File Transfer
  - **fileServer** = HTTP Server (any file)
- Remote access, subsetting CDM files
  - **dodsC** = OPeNDAP (any CDM file)
  - **wms** = Web Map Server (grids)
  - **wcs** = Web Coverage Server (grids)
  - **ncss** = NetCDF Subset Service (grids)
  - **admin** = Administration/debug interface

Note, each server can change the service name in the xml catalogue.

e.g. http://motherlode.ucar.edu:8080/thredds/wcs/...

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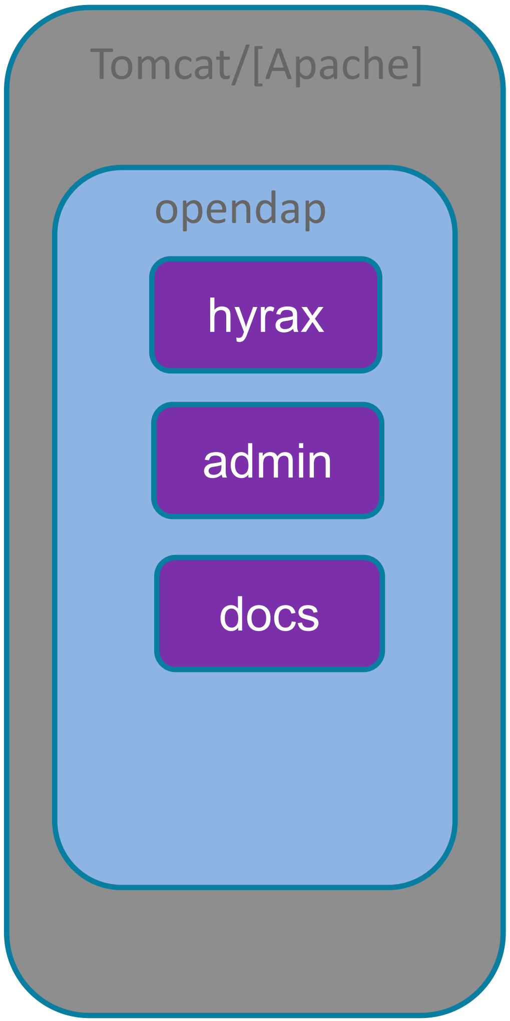
Hyrax service syntax



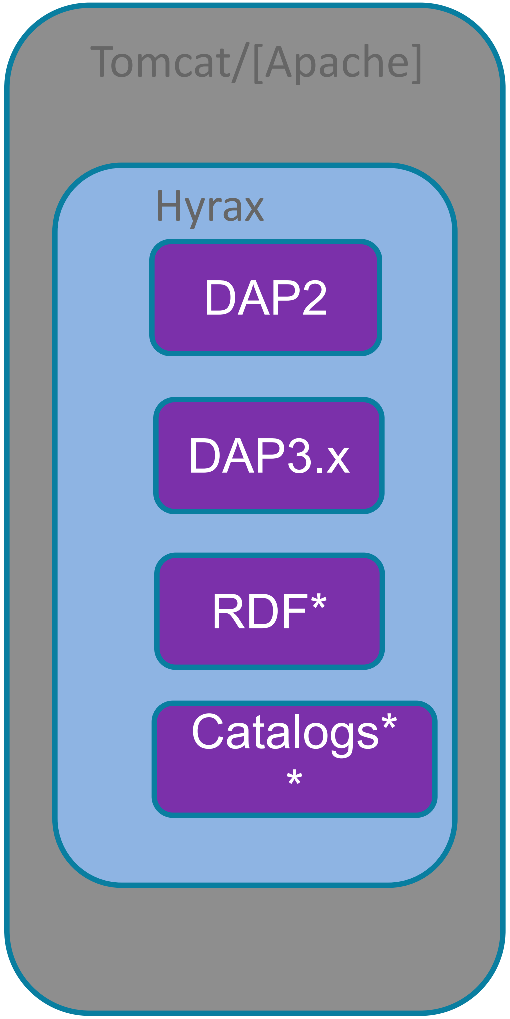
http://{server:port}/{contextPath}/{service}/…

{contextPath} = “opendap” (servlet default name)
{service} = “hyrax” | “admin” | “docs”
hyrax = catalog interface
admin = administration interface (v1.8+)
docs = documentation (v1.8+)

Note, each server can change the service name within the server configuration file.

e.g. http://test.opendap.org/opendap/hyrax/…

Hyrax Data Service



- DAP2 and DAP3.x as the protocol develops
- Other dataset responses*
  - ASCII & NetCDF renderings of data (not limited to data natively stored in netCDF)
  - RDF
  - ISO 19115 and the conformance rubric (Hyrax 1.8)
- Other server responses**
  - THREDDS catalogs

Note: Hyrax and TDS are not mutually excusive; Sites can install both with little extra effort.

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THREDDS Catalog Service

Catalogue contains:

- Hierarchy of logical views of data sets in files
    - The file path is hidden from the user, so the data move without breaking the URL.
- List of services which can operate on data sets
    - DAP service
    - HTTP service
    - Other services
- List of file's attributes and metadata contents

Note that THREDDS catalogs are separate from either TDS or Hyrax; both implement them differently, but to clients they have the same information in both servers

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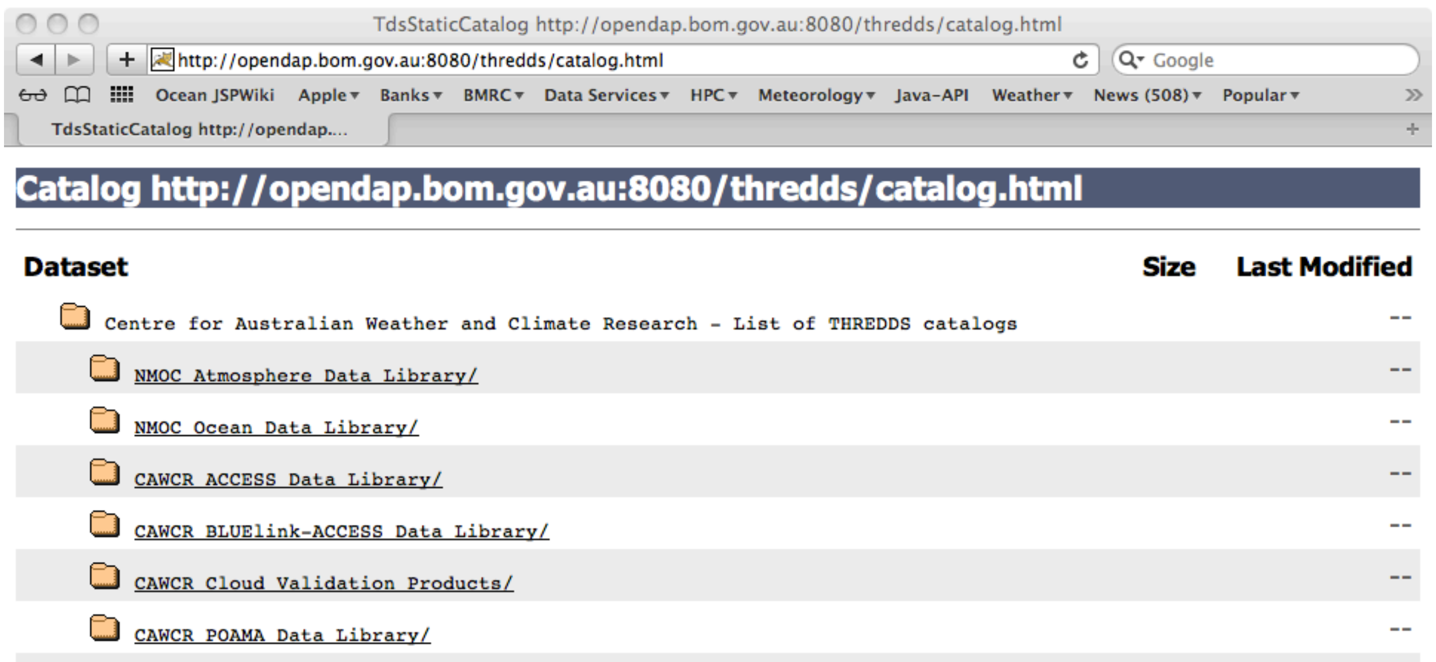
CAWCR Research Data Server

- Location: http://opendap.bom.gov.au:8080/thredds
- Unidata THREDDS Data Server v4.2.8
  - http://www.unidata.ucar.edu/projects/THREDDS/tech/TDS.html
  - The THREDDS Data Server (TDS) is a JavaSevlet, and is contained in a single war file, which allows very easy installation into Tomcat web server.



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Data Sets for Research

NMOC ACCESS catalog

- ACCESS-G
- ACCESS-R
- ACCESS-A
- ACCESS-T
- ACCESS-TC
- ACCESS-AD
- ACCESS-BN
- ACCESS-PH
- ACCESS-SY
- ACCESS-VT

NMOC Ocean catalog

- OceanMAPS v1
- OceanMAPS v2
- WW3
- GHRSST

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Tutorial: Explore the Catalogs

- View the data catalogs at the following services
  - Can you tell the difference between the TDS and the Hyrax data service?
  - Can you find the GHRSST sea temperature data sets for Global ¼ degree resolution?

- Data servers located at
  - Workshop (http://192.168.1.1:8090/opendap)
  - Bureau of Meteorology (http://opendap.bom.gov.au:8080/thredds)
  - CSIRO (http://opendap.csiro.au/thredds)
  - ANU/NCI (http://opendap.nci.org/thredds)
  - OPeNDAP, inc (http://test.opendap.org/opendap)
  - TPAC (http://opendap-tpac.arcs.org.au/thredds)

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DAP Responses

DAP2 defines three response types:

- **DAS**: A text document that contains data set attributes
- **DDS**: A text document that contains data set variable types and names
- **DODS**: A quasi-multipart MIME document that contains the DDS and associated binary values for a data request

DAP3.x defines two additional response types:

- **DDX**: An XML document that combines both variable type and name information along with attributes
- **DataDDX**: A multipart MIME document that combines a DDX with the associated binary values for a data request

TDS and Hyrax both support DAP2; Hyrax includes support for DAP3, TDS has support for the DDX

Tutorial: Explore the Catalog Responses

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List of the catalog contents (e.g. html source)

- http://opendap.bom.gov.au:8080/thredds/dodsC/gamssa_4deg/2011/catalog.html

List of the catalog contents (e.g xml source)

- http://opendap.bom.gov.au:8080/thredds/dodsC/gamssa_4deg/2011/catalog.xml

Select a file and view the catalog information (.e.g html source)

- http://opendap.bom.gov.au:8080/thredds/dodsC/gamssa_4deg/2011/catalog.html?dataset=gamssa_4deg/2011/20111106-ABOM-L4LRfnd-GLOB-v01-fv01.nc

View the catalog information, return the XML source, can you guess the URL?

Decipher the URL

http://opendap.bom.gov.au:8080/thredds/dodsC/gamssa_4deg/2011/20111106-ABOM-L4LRfnd-GLOB-v01-fv01.nc.ascii?lon[0:1:1439]

Given the OPeNDAP data request above, decipher the URL.

- Request Protocol? http
- Host name:port? //opendap.bom.gov.au:8080/
- ContextPath? thredds/
- Service? dodsC/
- Unique path to data set? gamssa_4deg/2009/
- Data reference? 20111106-ABOM-L4LRfnd-GLOB-v01-fv01.nc
- Return type? ascii
- Return variables? ?lon
- Return variable indice range? [0:1:1439] --> [start:skip:end]

Tutorial – Explore the DAP Responses

Given a OPeNDAP URL to a file,

- http://opendap.bom.gov.au:8080/thredds/dodsC/gamssa_4deg/2011/20111106-ABOM-L4LRfnd-GLOB-v01-fv01.nc

Find the OPeNDAP form by adding a ".html" to the above URI

- http://opendap.bom.gov.au:8080/thredds/dodsC/gamssa_4deg/2011/20111106-ABOM-L4LRfnd-GLOB-v01-fv01.nc.html

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Tutorial: .das, .dds, .ddx response

Find the OPeNDAP by adding a ".das" to the end of the URI

- http://opendap.bom.gov.au:8080/thredds/dodsC/gamssa_4deg/2011/20111106-ABOM-L4LRfnd-GLOB-v01-fv01.nc.das

Find the OPeNDAP by adding a ".dds" to the end of the URI

- http://opendap.bom.gov.au:8080/thredds/dodsC/gamssa_4deg/2011/20111106-ABOM-L4LRfnd-GLOB-v01-fv01.nc.dds

Find the OPeNDAP by adding a ".ddx" to the end of the URI

- http://opendap.bom.gov.au:8080/thredds/dodsC/gamssa_4deg/2011/20111106-ABOM-L4LRfnd-GLOB-v01-fv01.nc.ddx

Find the OPeNDAP by adding a ".help" to the end of the URI

- http://opendap.bom.gov.au:8080/thredds/dodsC/gamssa_4deg/2011/20111106-ABOM-L4LRfnd-GLOB-v01-fv01.nc.help

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Tutorial – other DAP responses

Look at:

- The other response listed on the .help page
  - Version: Server version from any ‘terminus’
  - HTML: A web interface for data which also helps ‘build URLs’
  - RDF: Metadata as triples; used with web-based reasoning systems

Constraints:

- Server-side functions
  - grid(): grid(analysed_sst,"-73.5<lon<-73","40<lat<40.5","time=946728000")
  - Call a function with no arguments… see what happens
  - linear_scale(): linear_scale(grid(…))
  - Use this URL for some experimentation:

    http://podaac-opendap.jpl.nasa.gov/opendap/allData/ghrsst/data/L4/GLOB/JPL_OUROCEAN/G1SST/

Tutorial: .ascii response

Find the OPeNDAP by adding a “.html” to the end of the URI

- http://opendap.bom.gov.au:8080/thredds/dodsC/gamssa_4deg/2011/20111106-ABOM-L4LRfnd-GLOB-v01-fv01.nc.html
- Click the box next to Variable: “lon”, observe the change to Data URL
  - …-fv01.nc?lon[0:1:1439]
- then click the button “Get ASCII” to see data returned.

- “.ascii” tells the OPeNDAP service to return the data in ASCII format.
  - http://opendap.bom.gov.au:8080/thredds/dodsC/gamssa_4deg/2011/20111106-ABOM-L4LRfnd-GLOB-v01-fv01.nc.ascii?lon

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LIST OF OPENDAP SERVERS

- OPeNDAP servers located at
  - Workshop (http://192.168.1.1:8090/hyrax)
  - Bureau of Meteorology (http://opendap.bom.gov.au:8080/thredds)
  - CSIRO (http://opendap.csiro.au/thredds)
  - ANU/NCI (http://opendap.nci.org/thredds)
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